

A Project report on

IOT-BASED SMART INTRUDER DETECTION SYSTEM

Submitted in partial fulfilment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

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Computer Science and Engineering (Data Science)

**SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY
(AUTONOMOUS)**

Rotarypuram Village, B K Samudram Mandal, Ananthapuramu - 515701

2025-2026

SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY

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Rotarypuram Village, BK Samudram Mandal, Ananthapuramu-515701

COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)



Certificate

This is to certify that the project report entitled **IoT-Based Smart Intruder Detection System** is the bonafide work carried out by **Sravani P, Vyshnavi G, Siraj Ahammad C, Sohith Kumar Reddy B** bearing Roll Number **234G5A3210, 224G1A32B9, 224G1A3290, 224G1A3292** in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology** in **Computer Science and Engineering (Data Science)** during the academic year 2025-2026.

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DECLARATION CERTIFICATE

We, students of **Computer Science and Engineering (Data Science)**, **SRINIVASA RAMANUJAN INSTITUTE OF TECHNOLOGY(AUTONOMOUS)**, Rotarypuram, hereby declare that the dissertation entitled **“IoT-Based Smart Intruder Detection System”** embodies the report of our project work carried out by us during IV year under the guidance of Dr. D Rajesh Babu , M.Tech., Ph.D, Assistant Professor, Department of Computer Science and Engineering, Srinivasa Ramanujan Institute of Technology, and this work has been submitted for the partial fulfilment of the requirements for the award of degree of Bachelor of Technology.

The results embodied in this project report have not been submitted to any other University or Institute for the award of any Degree or Diploma.

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ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful completion of any task would be incomplete without the mention of people who made it possible, whose constant guidance and encouragement crowned our efforts with success. It is a pleasant aspect that we now have the opportunity to express our gratitude for all of them.

It is with immense pleasure that we would like to express our indebted gratitude to our Guide, **Dr D Rajesh Babu, MTech, Ph.D., Associate Professor, Computer Science and Engineering**, who has guided us a lot and encouraged us in every step of the project work. We thank him for the stimulating guidance, constant encouragement and constructive criticism which have made it possible to bring out this project work.

We express our deep-felt gratitude to **Ms P Sirisha, Assistant Professor, Computer Science and Engineering (Data Science)** project coordinator, for her valuable guidance and unstinting encouragement, which enabled us to accomplish our project in time.

We are very much thankful to **Dr P Chitralingappa, M.Tech., Ph.D., Associate Professor & Head of the Department, Computer Science and Engineering (Data Science)**, for his kind support and for providing necessary facilities to carry out the work.

We wish to convey our special thanks to **Dr G Bala Krishna, Ph.D., Principal of Srinivasa Ramanujan Institute of Technology (Autonomous)**, for giving the required information in doing our project work. Not to forget, we thank all other faculty and non-teaching staff, and our friends who have directly or indirectly helped and supported us in completing our project in time.

We also express our sincere thanks to the Management for providing excellent facilities.

Finally, we wish to convey our gratitude to our families who fostered all the requirements and facilities that we need.

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ABSTRACT

The Smart Intruder Detection System is designed to provide an intelligent and reliable security solution by integrating multiple sensors with a Raspberry Pi controller. The system continuously monitors the surroundings using a PIR sensor to detect human motion, an ultrasonic sensor to identify vehicle movement, and a sound sensor to capture unusual acoustic activities. A USB camera is used for real-time face detection when motion or abnormal activity is detected. Upon identifying a potential intruder, the system triggers visual and audio alerts through LEDs and a buzzer, ensuring immediate indication of security threats. By combining sensor data processing with image-based monitoring, the proposed system enhances accuracy, reduces false alarms, and offers a cost-effective, scalable approach for modern security applications in homes, offices, and restricted areas.

The proposed system integrates a Raspberry Pi controller with multiple sensors, including a Passive Infrared (PIR) sensor for motion detection, an ultrasonic sensor for object movement detection, and a sound sensor for detecting unusual acoustic activities. These sensors continuously monitor the environment and transmit data to the Raspberry Pi for processing. Upon detection of any suspicious activity, the system activates a USB camera to capture images, which are further processed using OpenCV for face detection.

The system is designed to generate real-time alerts by activating LEDs and a buzzer while simultaneously sending email notifications with captured images to authorized users through the Simple Mail Transfer Protocol (SMTP). This ensures immediate response and effective monitoring without continuous human supervision.

The integration of multiple sensors improves detection accuracy and minimizes false alarms. The system is cost-effective, scalable, and suitable for applications in homes, offices, warehouses, and other secured environments. Overall, the project demonstrates the effective use of IoT and embedded systems in developing a reliable and intelligent security solution.

Keywords: Intruder Detection ,Face Recognition ,Motion Detection , Raspberry Pi , Sound Detection, Real-Time Monitoring.

PROJECT CONTRIBUTION TOWARDS SUSTAINABLE DEVELOPMENT GOALS

Project Title:

IoT-Based Smart Intruder Detection System

SDG - Project Mapping:

S.NO	Sustainable Development Goals	Mapped (Yes/No)
1	SDG 1: No Poverty	No
2	SDG 2: Zero Hunger	No
3	SDG 3: Good Health and Well-Being	Yes
4	SDG 4: Quality Education	No
5	SDG 5: Gender Equality	No
6	SDG 6: Clean Water and Sanitation	No
7	SDG 7: Affordable and Clean Energy	No
8	SDG 8: Decent Work and Economic Growth	Yes
9	SDG 9: Industry, Innovation and Infrastructure	Yes
10	SDG 10: Reduced Inequalities	No
11	SDG 11: Sustainable Cities and Communities	Yes
12	SDG 12: Responsible Consumption and Production	No
13	SDG 13: Climate Action	No
14	SDG 14: Life Below Water	No
15	SDG 15: Life on Land	No
16	SDG 16: Peace, Justice and Strong Institutions	Yes
17	SDG 17: Partnerships for the Goals	No

Explanation:

- Enhances security by detecting unauthorized entry in real time.
- Reduces crime and improves safety in homes, offices, and restricted areas.
- Supports smart surveillance and automated monitoring systems.
- Helps authorities and individuals respond quickly to security threats.

Expected Impact

Parameter	Impact
Security	Improved protection against intrusions
Safety	Reduction in thefts and unauthorized access
System Efficiency	Real-time detection and alert system
Infrastructure	Supports smart homes and smart city security

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LIST OF ABBREVIATIONS

IoT	Internet of Things
SoC	System on Chip
GPIO	General Purpose input/output
DSI	Display Serial Interface
GSM	Global System for Mobile Communication
Wi-Fi	Wireless Fidelity
MIPI	Mobile Industry Processor Interface
CSI	Camera Serial Interface
JTAG	Joint Test Action Group
TAP	Test Access Port
HDMI	High Definition Multimedia Interface
SIM800L	GSM Communication Module
GPIO	General Purpose Input Output
IDE	Integrated Development Environment
VS Code	Visual Studio Code

CHAPTER - 1

INTRODUCTION

Security is one of the most important concerns in modern society due to the increasing number of thefts, unauthorized entries, and security breaches. Traditional security systems such as basic alarms and CCTV cameras often require continuous human monitoring and may not work effectively in all situations, especially in low-light or dark environments where cameras may fail to capture clear evidence and gives the false alerts. Iot-Based Smart Intruder Detection System is designed to provide a reliable and automated security solution by integrating multiple sensors with a Raspberry Pi controller. The system continuously monitors the environment using motion sensors, sound sensors, and ultrasonic sensors to detect unusual activities. When suspicious activity is detected, the system automatically activates alerts and captures images using a USB camera. This project mainly focuses on detecting intruders in low-light and dark conditions in restricted areas such as warehouses, offices, banks, and secure buildings. By using sensors that do not rely completely on visible light, the system can identify motion and suspicious activity even in poor lighting conditions. The combination of multiple detection techniques, intelligent processing, and automated alerts improves the efficiency of traditional security systems while reducing false alarms, providing a reliable and cost-effective solution for modern security needs.

1.1 Problem Statement

In many restricted areas such as warehouses, offices, and security zones, intruders often attempt unauthorized entry during night time or in poorly lit environments where visibility is limited. Traditional surveillance systems mainly rely on cameras, which may not perform effectively in low-light or dark conditions, resulting in unclear footage or failure to detect intruders. Additionally, these systems depend heavily on manual monitoring, where security personnel must continuously watch camera feeds, making the process time-consuming and inefficient. Systems that rely on a single sensor also tend to generate false alarms due to environmental factors such as animals, wind, or noise. Therefore, there is a need for a smart and automated intruder detection system that can

operate efficiently even in low-light and dark conditions by integrating multiple sensors to detect motion, sound, and object movement simultaneously and automatically generate alerts when suspicious activity is detected.

1.2 Objectives of the Project

- To develop IoT Based Smart Intruder Detection System using Raspberry Pi.
- To detect unauthorized individuals even in dark or low-light environments and during abnormal conditions.
- The system aims to minimize false alarms by integrating multi-sensor data fusion, face recognition, and anomaly detection techniques.
- The system aims to intelligently differentiate between humans, animals, shadows, and other non-threatening movements by combining multi-sensor data with image-based analysis.
- Another key objective is to implement real-time face recognition and sound detection to enhance security verification using USB camera.
- The system is designed to provide immediate alerts when an intruder or abnormal activity is detected. It automatically captures images using the USB camera and sends them to authorized users through email notifications for quick response.
- The system is designed to provide immediate alerts when an intruder or abnormal activity is detected through the email notifications along with captured coloured photos even in the low or dark conditions .
- It activates LEDs and a buzzer to generate visual and audio alerts, ensuring real-time indication of potential security threats.

1.3 Scope of the Project

The scope of the Smart Intruder Detection System is to develop and implement a hardware-based security framework that enables real-time monitoring and detection of unauthorized entry in restricted areas, particularly under low-light and dark conditions. The project focuses on integrating a Raspberry Pi controller with multiple sensors such

as PIR sensors for human motion detection, ultrasonic sensors for movement detection, and sound sensors for identifying unusual acoustic activities. A USB camera is incorporated to capture colored images whenever suspicious activity or a potential threat is detected. These captured images are automatically sent to authorized users through email notifications, while alert mechanisms such as LEDs and a buzzer provide immediate visual and audio warnings.

The system is designed for use in homes, offices, warehouses, banks, and other restricted environments where continuous security monitoring is required. By combining multi-sensor data processing with image capture and automated alert systems, the solution improves detection accuracy and reduces false alarms compared to traditional single-sensor security systems. While the current implementation focuses on sensor-based intrusion detection and email alert notifications, the framework can be further extended in the future by integrating advanced technologies such as artificial intelligence-based face recognition, mobile application notifications, cloud storage, and remote surveillance systems. The project primarily addresses automated intrusion detection, real-time monitoring, and enhanced security management for restricted areas.

CHAPTER - 2

LITERATURE SURVEY

The literature on smart security and intruder detection systems highlights the growing need for automated surveillance solutions capable of monitoring restricted areas effectively, especially during low-light and nighttime conditions. Many studies emphasize the use of sensor-based technologies such as PIR sensors, ultrasonic sensors, and sound sensors to detect human motion, object movement, and abnormal sound activities. Traditional security systems mainly rely on CCTV cameras, which often require continuous human monitoring and may not function effectively in poor lighting conditions. Researchers have proposed integrating embedded platforms such as Raspberry Pi with multiple sensors and cameras to improve detection accuracy and reduce false alarms. In addition, several studies show that combining sensor data with image processing techniques can enhance the reliability of intrusion detection systems. These systems aim to provide real-time monitoring, automated alerts, and improved security management by detecting suspicious activities and notifying authorized users immediately.

2.1 Review of Previous Work

- **MS. Mamatha , P. Aakash, et al. " Intrusion Detection for Smart Home AlarmSecurity System " [1].**
 - This study presents a cost-effective smart home security system developed using Raspberry Pi, integrating motion detection, real-time image capture, SMS and email alerts, and automated door locking. The system utilizes OpenCV for motion tracking and a GSM module for communication, ensuring quick alert delivery and continuous operation using a backup power supply. The research demonstrates high detection accuracy and highlights the system's scalability and affordability. This work supports the development of efficient Raspberry Pi-based security systems similar to the proposed intruder detection model.

- **Pawan Game, Pranjali Gangurde , et al. " Intruder Detection System Using Machine Learning " [2].**
 - This paper proposes an intelligent surveillance system that integrates motion detection, face recognition, and face mask detection using machine learning and computer vision techniques. The system enables real-time monitoring, identification, and alert generation, improving overall security and safety. It provides a scalable solution suitable for various environments such as public spaces and organizations. The study supports the use of advanced algorithms and real-time processing in enhancing intruder detection systems.
- **Muhammad Sajid, Kaleem Razzaq Malik, et al. " Enhancing intrusion Detection: a hybrid machine and deep learning approach" [3].**
 - This research focuses on improving intrusion detection in network systems using a hybrid model that combines machine learning and deep learning techniques. The model utilizes algorithms such as XGBoost, CNN, and LSTM for feature extraction and classification, achieving high accuracy and low false acceptance rates. The study highlights the importance of advanced detection techniques in handling large-scale data and evolving threats. This work emphasizes the role of intelligent algorithms in improving detection accuracy, which is relevant to modern security systems.
- **Hiba Hameed Ali, Jolan, Rakan Naif , et al. " A New Smart Home Intruder Detection System Based on Deep Learning " [4].**
 - This study introduces a smart home security system based on deep learning for face recognition using the EfficientNet-B4 model. The system is optimized for implementation on Raspberry Pi using transfer learning and lightweight model conversion. It achieves high accuracy even under varying lighting conditions and facial expressions. The research highlights the effectiveness of deep learning in enhancing face recognition performance, supporting the integration of intelligent image processing in intruder detection systems.

- **Menaga, Priyadharshini, et al. " A Smart Intruder Detection System " [5].**
 - This paper presents an intruder detection system using PIR sensors and a Raspberry Pi camera to detect and capture images of intruders. The system uses OpenCV-based face recognition to compare captured images with stored authorized user data. If an unauthorized person is detected, the system sends an alert along with the image to the user via email. The study demonstrates an effective approach for real-time intruder detection and notification, forming a foundation for automated security systems.

Sr No.	Paper title	Technologies used	Strengths	Weakness
1	"e-KTP as the basis of home security system using arduinoUNO"	RFID	i) Only unique ID number can give access to open door ii) Web based logging system	i) Metals and liquids can impact signal
2	"Intelligent acoustic and vibration recognition/alert systems for security breaching detection, close proximity danger identification, and perimeter protection"	Acoustic and Vibration Recognition	i) seismic analyzer which discriminates between human footsteps and other seismic events ii) False signal detection	i) Practically not accurate
3	"Low power accelerometer based intrusion and tamper detector"	Vibration analysis using Accelerometer	i) Cost effective ii) Effective vibration sensing iii) Low power consumption	i) Implementation is difficult
4	"IoT based Intruder Detection System Using GSM"	GSM	i) Alert warning to owner ii) Simple and easy implementation	i) Not efficient ii) Not much reliable
5	"Fuzzy logic based method to estimate the risk of alarm system false detection,"	Fuzzy logic	i) Accurate sensing of data ii) Environmental signal and noise can be distinguished	i) Special sensors required ii) Expensive
6	"Identification of Damaging Activities for Perimeter Security"	Support Vector Machine (ML)	i) Feature extraction ii) Accurate data analysis	i) Improvement for incremental learning ability
7	"Home security system using internet of things"	Arduino UNO	i) Open source ii) Easy implementation	i) Not reliable ii) Not secure

8	Securing IoT Devices and Securely Connecting the Dots Using REST API and Middleware	Rest API	i) End to end encryption ii) Secure IOT devices	i) Resource constraints
9	Automatic Service Request System for Security in Smart Home Using IoT	Raspberry PI	i) Multipurpose use such as security, fire safety etc	i) More complex ii) Expensive
10	A Smart IoT Security System for Smart-Home Using Motion Detection and Facial Recognition	Motion detection Facial recognition	i) High degree of classification ii) Accurate security sensing	i) Low processing power ii) Huge database required
11	Security and Privacy of Smart Home Systems Based on the Internet of Things and Stereo Matching Algorithms	Stereo matching algorithms	i) Highly secured network ii) Easy home automation and security	i) Bulky ii) Expensive
12	Theft Detection System using PIR Sensor	PIR sensor	i) Real time video sensing ii) Direct surveillance	i) Precautions measures by owner
13	Ambient Intelligence and IoT Based Decision Support System for Intruder Detection	Face detection	Accurate human face sensing	Not useful for non-human security breaches
14	IoT Based Mobile Smart Home Surveillance Application	NodeMCU Wifi module	Smart Home monitor	i) Lack of database ii) Less sensors iii) Not reliable for security options
15	IoT based Smart Home Surveillance and Automation	GSM, Raspberry PI	Home Security Surveillance System	i) Easy to attack ii) Data leakage

Table 2.1 : Literature Survey

CHAPTER - 3

PROJECT ANALYSIS

3.1 Introduction

Planning for this project involves a systematic approach to designing and implementing a smart intruder detection system capable of monitoring restricted areas and identifying unauthorized access, especially under low-light and dark conditions. The planning phase focuses on identifying the required hardware components such as Raspberry Pi, PIR sensor, ultrasonic sensor, sound sensor, USB camera, LEDs, and buzzer, along with defining the software architecture that enables sensor data processing, image capture, and alert generation. Special attention is given to ensuring accurate motion detection, reliable sensor data acquisition, and immediate response mechanisms when suspicious activity is detected. The system design emphasizes reliability, cost-effectiveness, and continuous monitoring of restricted environments such as offices, warehouses, and security zones. A structured development plan is established that includes hardware integration, sensor calibration, camera configuration, software programming using Python, system testing, and deployment. This coordinated approach ensures smooth interaction between hardware and software components, enabling efficient intruder detection and real-time alert generation.

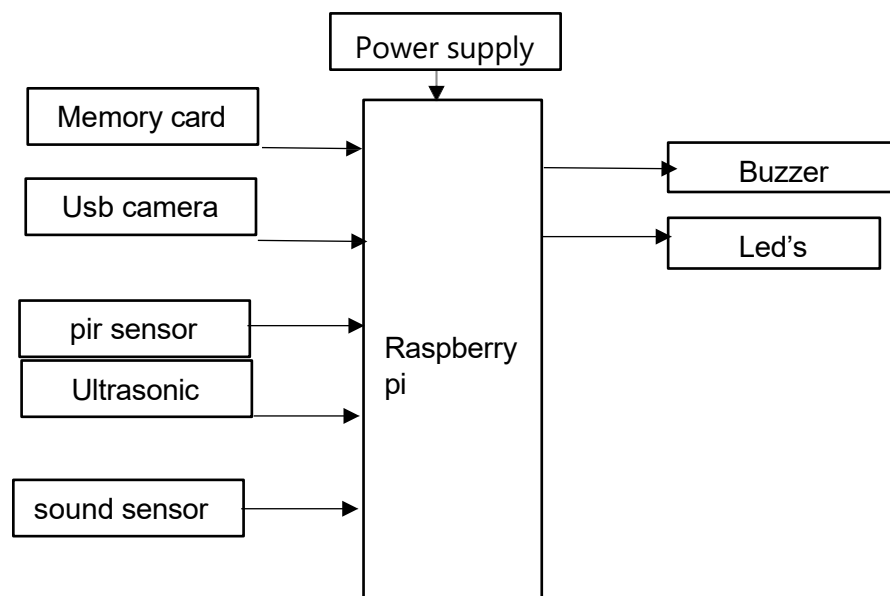


Fig.3.1 Planning

3.1.1 Project Planning:

The first stage of project planning involves requirement analysis and system objective identification. During this phase, the main goal of the system is defined, which is to detect intruders in restricted areas, particularly in low-light or dark conditions. The key functionalities of the system include motion detection using PIR sensors, object movement detection using ultrasonic sensors, sound detection using sound sensors, image capture using a USB camera, and alert notification through LEDs, buzzers, and email notifications.

The second stage involves system design and architecture planning. In this phase, appropriate hardware components such as Raspberry Pi, PIR sensor, ultrasonic sensor, sound sensor, and USB camera are selected. The software architecture is designed using Python programming to process sensor data and control the camera and alert mechanisms. The system workflow is planned to ensure that whenever suspicious activity is detected, the system captures images and immediately sends alerts to authorized users.

The final stage focuses on development, testing, and deployment. Hardware components are connected and tested to ensure proper functionality and accurate sensor readings. The software is developed to monitor sensor inputs continuously and trigger alerts when an intrusion is detected. Comprehensive testing is performed to verify motion detection accuracy, camera image capture, email alert functionality, and system reliability before final deployment.

3.2 Introduction to Existing System

The existing security systems commonly rely on CCTV cameras and simple motion detectors to monitor restricted areas. These systems capture video footage and store it for later analysis. However, many of these systems require continuous human monitoring to identify suspicious activities.

In addition, traditional surveillance systems often perform poorly in low-light or dark environments without additional lighting equipment. Systems that rely on a single sensor may also generate false alarms due to environmental disturbances such as animals, wind, or noise. Furthermore, many existing systems lack intelligent processing capabilities and real-time alert mechanisms. As a result, security personnel may not

receive immediate notifications when an intrusion occurs. These limitations highlight the need for a smart and automated intruder detection system that can provide reliable monitoring and instant alerts.

3.2.1 Disadvantages of Existing System:

- **Poor Performance in Low-Light and dark Conditions:** the existing system mainly depend on cameras that perform poorly in dark environments. This reduces their ability to detect intruders effectively at night.
- **Lack of Intelligent Anomaly Detection:** Existing systems cannot differentiate between humans, animals, shadows, or other non-threatening movements. They lack multi-sensor data fusion and image-based analysis, which reduces detection accuracy.
- **Lack of Face Recognition and Sound Detection:** Existing security systems do not support real-time face recognition or sound detection. This makes it difficult to verify identities and detect unusual acoustic activities or suspicious voices.
- **Absence of Instant Alert and Notification:** existing systems will not provide automatic alerts or real-time notifications in night times or dark conditions . This can delay the response to potential security threats.
- **High False Alarm Rate:** Single-sensor systems may trigger alarms due to environmental disturbances such as animals, wind, or noise.

3.3 Proposed System

The proposed Smart Intruder Detection System introduces an automated security monitoring framework that integrates multiple sensors with a Raspberry Pi controller. The system continuously monitors the environment using PIR sensors for motion detection, ultrasonic sensors for movement detection, and sound sensors for detecting unusual sound activities.

When any suspicious activity is detected, the Raspberry Pi processes the sensor data and activates the USB camera to capture colored images of the surrounding environment. These images are then automatically sent to authorized users through email notifications to provide immediate information about potential threats.

The system also activates alert mechanisms such as LEDs and a buzzer to provide visual and audio warnings in real time. This multi-sensor approach improves detection accuracy, reduces false alarms, and enhances security monitoring in restricted areas.

3.3.1 Advantages of Proposed System:

- **Effective Intruder Detection in Low-Light Conditions:** The system uses sensors that can detect motion and activity even in dark environments.
- **Real-Time Monitoring and Alerts:** Immediate alerts are generated through LEDs, buzzers, and email notifications when suspicious activity is detected.
- **Multi-Sensor Detection:** Combining PIR, ultrasonic, and sound sensors improves detection accuracy and reduces false alarms.
- **Automated Image Capture:** The USB camera captures colored images of intruders and sends them to authorized users via email.
- **Anomaly Detection and Intelligent Differentiation:** The system aims to intelligently differentiate between humans, animals, shadows, and other non-threatening movements by combining multi-sensor data with image-based analysis.
- **Reduction of False Alarms:** The system aims to minimize false alarms by integrating multi-sensor data fusion, face recognition, and anomaly detection techniques.

CHAPTER - 4

REQUIREMENTS

The Smart Intruder Detection System requires software components for sensor monitoring, image processing, and alert generation. The system is developed using Python programming language on the Raspberry Pi platform. Python is used to interface with sensors, process captured images, and manage alert mechanisms. The system also uses OpenCV for image processing and face detection. This library allows the system to capture images from the USB camera and perform face detection when motion or suspicious activity is detected. Additional Python libraries such as GPIO, NumPy, and SMTP are used to control hardware components and send email notifications. The operating system used for the Raspberry Pi is Raspberry Pi OS (Linux-based), which provides support for hardware interfacing and Python-based applications.

4.1 Requirements Specification

The system requires both functional and non-functional specifications to ensure smooth operation. Functionally, the system must detect motion using a PIR sensor, detect object movement using an ultrasonic sensor, and detect unusual sounds using a sound sensor. When suspicious activity is detected, the system must capture images using a USB camera and analyze them for face detection. The system must also generate alerts using LEDs, a buzzer, and email notifications.

Non-functionally, the system must operate continuously, provide fast detection response, and minimize false alarms. It must be reliable, secure, and capable of working effectively in low-light or dark environments.

4.1.2 Software Requirements

Programming Language::

- Python for system control and processing
- Python libraries for sensor interfacing and automation
- Embedded C

Image Processing:

- OpenCV library for image capture and face detection

Development Tools:

- Visual Studio Code
- Arduino IDE

Communication Services:

- SMTP protocol for sending email alerts with captured images

4.1.3 Hardware Requirements:

- **Transformer:**

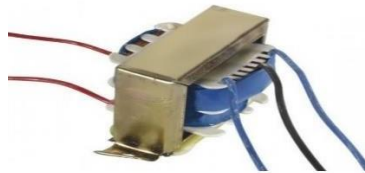


Fig 4.1: Transformer

A transformer is a static electrical gadget that exchanges control between at least two circuits. A fluctuating current creates a changing attractive motion in one transformer curl, which thus actuates a differing electromotive power over a second loop twisted around a similar center.

- **Raspberry Pi**



Fig 4.2: Raspberry Pi

The Raspberry Pi is a compact, credit-card-sized single-board computer developed by the Raspberry Pi Foundation in the United Kingdom to promote learning in computer science, programming, and electronics. It is widely used for educational and IoT-based applications due to its low cost, small size, and flexibility. The device is powered by supporting operating systems like Linux and programming languages such as Python, C, and Java. In this project, the Raspberry Pi acts as the central controller that connects and processes data from sensors like the PIR sensor, ultrasonic sensor, and sound sensor. It analyzes the inputs to detect suspicious activities, controls the USB camera for capturing images, and triggers alert mechanisms such as LEDs, buzzer, and email notifications. Thus, it plays a crucial role in enabling real-time monitoring, data processing, and automated intruder detection in the system.

	Model A	Model B	Model B+
Target price:	US\$25	US\$35	
SoC:	Broadcom BCM2835 (CPU, GPU, DSP, SDRAM, and single USB port)		
CPU:	700 MHz ARM1176JZF-S core (ARM11 family, ARMv6 instruction set)		
GPU:	Broadcom Video Core IV @ 250 MHz		
Memory	256 MB	512 MB (shared with GPU)	
USB 2.0 ports:	1 (direct from BCM2835 chip)	2(via the on-board 3-port USB hub)	4(via the on board 5-port USB hub)
Video input:	15-pin MIPI camera interface (CSI) connector, used with the Raspberry Pi		
Video outputs:	Composite RCA (PAL and NTSC) –in model B+ via 4-pole 3.5 mm jack, HDMI (rev 1.3 & 1.4), raw LCD Panels via DS		
Audio outputs:	3.5 mm jack, HDMI, and, as of revision 2 boards, I²S audio (also potentially		
Power	300 mA (1.5 W)	700 mA (3.5 W)	600 mA (3.0 W)
Power	5 V via Micro USB or GPIO header		
Size:	85.60 mm × 56 mm (3.370 in × 2.205 in) – not including protruding		
Weight:	45 g (1.6 oz)		

Table 4.1 : Specifications

- **WEB CAMERA**



Fig 4.3: Web Camera

A **webcam** is a [video camera](#) that feeds or [streams](#) an image or video in real time to or through a [computer network](#), such as the [Internet](#). Webcams are typically small cameras that sit on a desk, attach to a user's monitor, or are built into the hardware. Webcams can be used during a video chat session involving two or more people, with conversations that include live audio and video. Webcam software enables users to record a video or stream the video on the Internet. As video streaming over the Internet requires much [bandwidth](#), such streams usually use [compressed formats](#). The maximum resolution of a webcam is also lower than most handheld video cameras, as higher resolutions would be reduced during transmission. The lower resolution enables webcams to be relatively inexpensive compared to most video cameras, but the effect is adequate for video chat sessions.

- **PIR Sensor**



Fig 4.4: PIR Sensor

All living objects, whose body temperature is more than 0oC, emit the heat in form of infrared radiation through their body, also called as thermal radiations. This Radiated energy is invisible to human eye. These Signals can be detected by using PIR sensor which is specially designed for such purpose.

The PIR sensors are more complicated than the other sensors as they consists of two slots. These slots are made of a special material which is sensitive to IR. The Fresnel lens is used to see that the two slots of the PIR can see out past some distance. When the sensor is inactive, then the two slots sense the same amount of IR. The ambient amount radiates from the outdoors, walls or room, etc.

When a human body or any animal passes by, then it intercepts the first slot of the PIR sensor. This causes a positive differential change between the two bisects. When a human body leaves the sensing area, the sensor generates a negative differential change between the two bisects. The infrared sensor itself is housed in a hermetically sealed metal to improve humidity/temperature/noise/immunity. There is a window which is made of typically coated silicon material to protect the sensing element.

- **Ultrasonic Sensor**



Fig 4.5: Ultrasonic Sensor

An ultrasonic sensor transmit ultrasonic waves into the air and detects reflected waves from an object. There are many applications for ultrasonic sensors, such as in intrusion alarm systems, automatic door openers and backup sensors for automobiles.

Accompanied by the rapid development of information processing technology, new fields of application, such as factory automation equipment and car electronics, are increasing and should continue to do so. Using its unique piezoelectric ceramics manufacturing technology developed over many years, Murata has developed various

types of ultrasonic sensors which are compact and yet have very high performance. The information contained in this catalog will help you to make effective use of our ultrasonic sensors.

As shown above the HC-SR04 Ultrasonic (US) sensor is a 4 pin module, whose pin names are Vcc, Trigger, Echo and Ground respectively. This sensor is a very popular sensor used in many applications where measuring distance or sensing objects are required. The module has two eyes like projects in the front which forms the Ultrasonic transmitter and Receiver. The sensor works with the simple high school formula that

$$\text{Distance} = \text{Speed} \times \text{Time}$$

- **Sound Sensor**



Fig 4.6: Sound Sensor

The sound [sensor is one type of module](#) used to notice the sound. Generally, this module is used to detect the intensity of sound. The applications of this module mainly include switch, security, as well as [monitoring](#). The accuracy of this sensor can be changed for the ease of usage.

The working of this sensor is related to human ears. Because human eye includes a diaphragm and the main function of this [diaphragm](#) is, it uses the vibrations and changes into signals. Whereas in this sensor, it uses a microphone and the main function of this is, it uses the vibrations and changes into current otherwise voltage.

Generally, it includes a diaphragm which is designed with magnets that are twisted with metal wire. When sound signals hit the diaphragm, then magnets within the sensor vibrates & simultaneously current can be stimulated from the coils.

- **Buzzer**



Fig 4.7: Buzzer

A buzzer or beeper is an audio signaling device, which may be mechanical, electromechanical, or piezoelectric. Typical uses of buzzers and beepers include alarm devices, timers and confirmation of user input such as a mouse click or keystroke. Buzzer is an integrated structure of electronic transducers, DC power supply, widely used in computers, printers, copiers, alarms, electronic toys, automotive electronic equipment, telephones, timers and other electronic products for sound devices. Active buzzer 5V Rated power can be directly connected to a continuous sound, this section dedicated sensor expansion module and the board in combination, can complete a simple circuit design, to "plug and play."

Pin Number	Pin Name	Description
1	Positive	Identified by (+) symbol or longer terminal lead. Can be powered by 5V DC
2	Negative	Identified by short terminal lead. Typically connected to the ground of the circuit

Table 4.2 :Pin Configuration

Additional Components:

- Jumper wires
- Breadboard
- MicroSD card for Raspberry Pi OS

4.2 Functional Requirements

1. Motion Detection:

- The PIR sensor must detect human motion in the monitored area

2. Movement Detection:

- The ultrasonic sensor must detect object movement or presence

3. Sound Detection:

- The sound sensor must detect unusual sounds or suspicious acoustic activities

4. Image Detection:

- The USB camera must capture images when suspicious activity is detected

5. Face Detection:

- The system must perform face detection using OpenCV

6. Alert System:

- The system activate LED indicators and a buzzer when an intruder is detected

7. Email Notification:

- The system must send email alerts with captured images to authorized users

8. Continous Monitoring:

- The system must continuously monitor restricted areas for suspicious activity

4.3 Non-Functional Requirements

1. Performance:

- The system must detect motion and generate alerts within a few seconds.
- Sensor data and camera processing must operate in real time.

2. Scalability:

- The system should allow additional sensors or cameras to be integrated in the future.

3. Availability:

- The system must operate continuously for 24/7 surveillance.

4. Security:

- Access to system alerts and monitoring should be restricted to authorized users.

5. Reliability:

- The system must provide consistent sensor readings and stable monitoring.

6. Compatibility:

- The system must work with standard Raspberry Pi hardware and Python libraries.

4.4 Platform for Code Execution**Arduino IDE:**

The Arduino IDE software is an open source software, where we can have the example codes for the beginners. In the present world there are a lot of versions in the Arduino IDE in which present usage is Version 1.0.5. It is very easy to connect the PC with Arduino Board.

First we have to install the Arduino IDE software according to the below instructions:

- Insert the CD-ROM or PENDRIVE which contains the software and then copy the Setup File to your desired location.
- After copying, now click on the setup you will see a window shown below
- Click on NO, not this time. Then after NEXT

- Click On NO, not this time. Then after NEXT

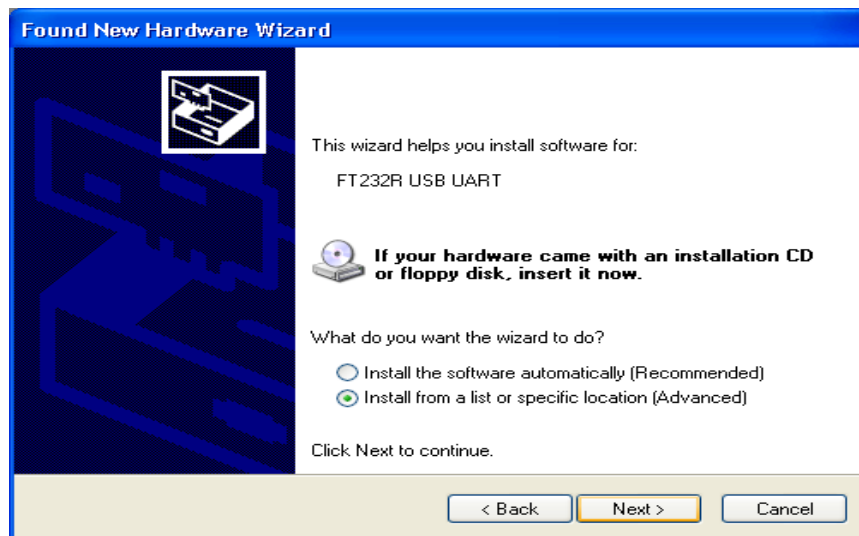


Fig 4.8: Arduino IDE Set Up

- Another Window opens –select Install from a list of specific location and NEXT
- Select “include this location in the search” and then click Browse option available in it
- Now it will Automatically check the USB driver and the software is installed click Finish



Fig 4.9: Arduino IDE Set Up

- Now click Finish, the Software will be downloaded.
- Now click on the Arduino IDE icon present on your Desktop. A window will

appear like this.

- For any sample programs, select FILE option→Examples.

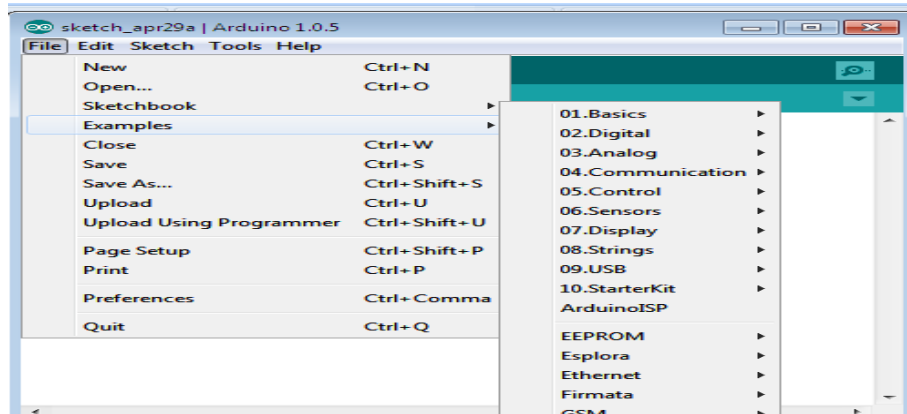


Fig 4.10: Arduino IDE Set Up

- After Entering the Sample Code in the file, it would look like this

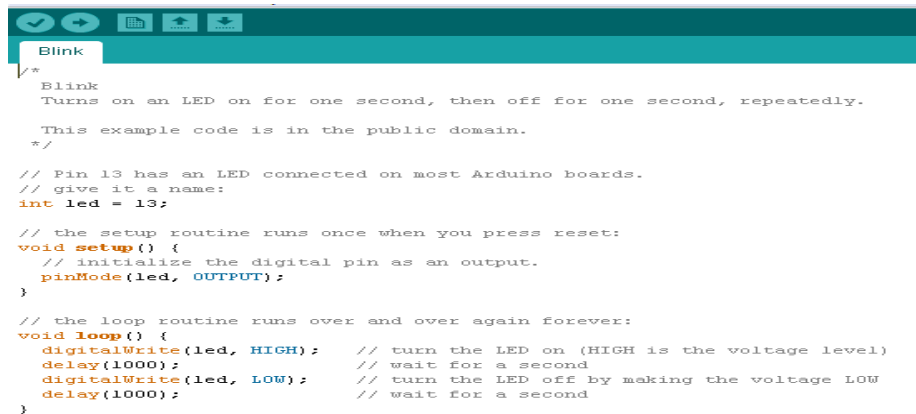


Fig 4.11: Sample Code

- Before Connecting we have to select which Board is used by the user, Basically UNO. By selecting TOOLS→Board→ARDUINO UNO
- Now to dump the in the board Connect the Arduino to the PC through the USB port available in it. Like this TOOLS→SERIAL PORT→COMM4,COMM8 etc;
- To verify the written Program select COMPILE option available in the software (✓).
- Now Connect the Board and select the COMM port and then UPLOAD the file in ARDUINO(→)
- To OPEN the Previous ARDUINO FILE selects (↑) option.
- To enter new files select NEW option.

- To Save the Existing File, Click on the(↓).
- To Send the Data Through Serial Monitor, Click on the (Q).

CHAPTER - 5

PROJECT ANALYSIS

5.1 Introduction

Analysis is an important phase in the development of the Smart Intruder Detection System, where the limitations of traditional security systems are studied and the requirements for an improved automated security solution are identified. The main purpose of this phase is to analyze how unauthorized entry can be detected effectively, especially in low-light or dark environments, and to design a system that can respond quickly to potential threats.

During the analysis stage, the requirements of the system are examined, including motion detection, sound detection, image capture, and alert generation. The system must be capable of monitoring restricted areas continuously using sensors such as PIR sensors, ultrasonic sensors, and sound sensors, while a USB camera captures images of suspicious activities. The Raspberry Pi acts as the central processing unit that integrates all sensors and processes the collected data.

Various constraints such as sensor accuracy, system cost, power requirements, and real-time response capability are also analyzed. The analysis phase also involves studying existing security systems such as CCTV surveillance and alarm systems to identify their limitations, including dependency on manual monitoring and poor performance in low-light conditions. The outcome of this phase includes system models such as block diagrams, data flow diagrams, and system architecture diagrams, which illustrate the interaction between hardware components and software modules. This analysis ensures that the proposed system provides a reliable and efficient security monitoring solution.

5.2 Phases in Analysis

- **Requirement Gathering:** This phase involves identifying the requirements for developing an effective security monitoring system. The system must detect human motion, unusual sounds, and object movement. It must also capture

images of intruders and generate alerts through LEDs, a buzzer, and email notifications.

- **Feasibility Study:** The technical feasibility is analyzed by evaluating the compatibility of Raspberry Pi with sensors such as PIR sensors, ultrasonic sensors, and sound sensors. Operational feasibility ensures that the system can operate continuously with minimal human intervention. Financial feasibility focuses on developing a low-cost security solution.
- **Requirements Documentation:** All functional and non-functional requirements are documented clearly. This includes sensor operation, image capture mechanisms, alert generation methods, and email notification systems. These documents serve as a guideline for the system development process.
- **System Modeling:** System models such as block diagrams, use case diagrams, and data flow diagrams (DFD) are created to visualize how different system components interact. These diagrams represent the flow of data between sensors, the Raspberry Pi controller, the camera module, and the alert system.
- **Validation and Verification:** The identified requirements are reviewed to ensure they meet the main objective of detecting intruders accurately, even in low-light environments. The system design is validated to confirm that all components work together effectively.
- **Preparation of Analysis Report:** A detailed analysis report is prepared summarizing the system requirements, feasibility results, system models, and constraints. This report serves as the foundation for the system design and implementation phase.

5.3 Analysed Points

- **Understanding the Problem:** The system addresses the lack of continuous monitoring and anomaly detection and may not perform well in low-light and dark environments. By these Problems it Produce false alarms.
- **Identifying Requirements:** Functional requirements include motion detection, sound detection, image capture, face detection, and alert generation.

Non-functional requirements include reliability, real-time monitoring capability, and reduced false alarm rates.

- **Feasibility Study:** Technical feasibility ensures the integration of Raspberry Pi with sensors, camera modules, and alert devices. Financial feasibility ensures the system remains cost-effective compared to advanced commercial security systems.
- **System Modeling:** System architecture diagrams and block diagrams are used to represent the interaction between sensors, Raspberry Pi, camera modules, and alert mechanisms.
- **Validation and Refinement:** The system requirements are reviewed and refined to ensure accurate detection, quick alert generation, and reliable performance in different environmental conditions before proceeding to the system design phase.

CHAPTER - 6

DESIGN

6.1 Introduction

Design Design is an important phase in the development of the Smart Intruder Detection System, where the analysed requirements are transformed into a structured framework for implementation. This phase defines the overall system architecture, hardware–software integration, data flow, and alert mechanisms required for effective security monitoring. The system design explains how different hardware components such as the Raspberry Pi, PIR sensor, ultrasonic sensor, sound sensor, USB camera, LED indicators, and buzzer interact with the software modules to detect intruders and generate alerts. The Raspberry Pi acts as the central controller that collects data from sensors, processes the information, and triggers actions when suspicious activity is detected. The design also includes image capturing and face detection using OpenCV, which helps identify intruders by capturing images through the USB camera. When unusual motion, sound, or object movement is detected, the system automatically captures images and sends email notifications with colored images to authorized users. A well-structured design ensures system reliability, faster response time, reduced false alarms, and efficient security monitoring, especially in low-light and dark environments.

6.2 Data Flow Diagram

A Data Flow Diagram (DFD) represents how data flows through the Smart Intruder Detection System. It illustrates the movement of data between sensors, processing units, and alert mechanisms within the system. The DFD helps in understanding how the system collects input data, processes it using the Raspberry Pi, and generates alerts when an intruder is detected.

The Level 0 DFD (Context Diagram) provides a high-level overview of the entire Intruder Detection System. In this level, the whole system is represented as a single process that interacts with external entities such as sensors, the USB camera, and the authorized user. Sensors like the PIR sensor, ultrasonic sensor, and sound sensor continuously monitor the environment and send data to the Raspberry Pi. When

suspicious activity is detected, the system processes the data and sends alerts along with captured images to the authorized user through email notification.

The Level 1 DFD provides a more detailed representation of the system by breaking the main process into smaller sub-processes. These sub-processes include motion detection using the PIR sensor, object movement detection using the ultrasonic sensor, and sound detection using the sound sensor. When any abnormal activity is detected, the system activates the USB camera to capture images.

After capturing the images, the system performs face detection using OpenCV to analyze the captured images and identify possible intruders. Once the detection process is completed, the system generates alerts through LED indicators and a buzzer. At the same time, an email notification with the captured images is sent to the authorized user.

These interconnected processes work together to ensure that the system continuously monitors restricted areas, detects suspicious activities, and provides immediate alerts to enhance the security and reliability of the intruder detection system.

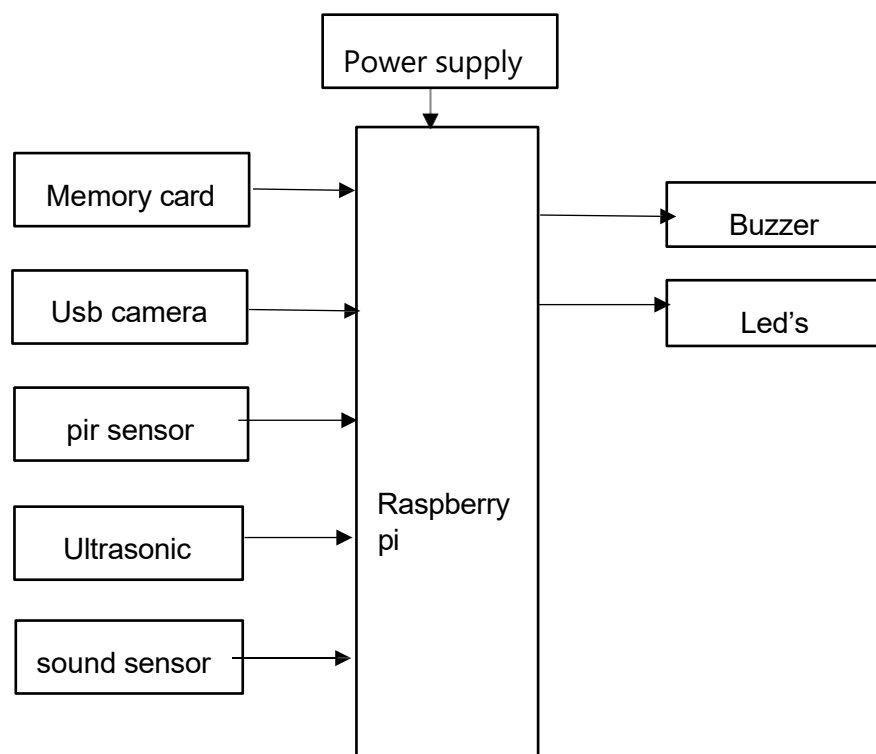


Fig 6.1: Data Flow Diagram

6.3 Software Model

Choosing an appropriate software development model is crucial for ensuring the efficiency, reliability, and maintainability of a project. The selection of a software model depends on various factors such as project complexity, requirements stability, available resources, and time constraints. Several software development models exist, including the Waterfall model, Agile methodology, Spiral model, V-Model, and Iterative model. Each model offers unique advantages and challenges, making it essential to analyse the project's requirements and goals before finalising the most suitable approach. The chosen model should align with the project's objectives, ensuring smooth development, testing, and deployment phases while minimising risks and enhancing adaptability to changes. Agile methodologies, such as Scrum and Kanban, enable teams to work in short iterations (sprints), delivering functional software at the end of each cycle. This model enhances collaboration between developers, stakeholders, and users, ensuring that the final product meets evolving expectations. The V-Model (Verification and Validation Model) extends the Waterfall approach by incorporating testing at every stage of development, ensuring that each phase is validated before proceeding. This model is particularly useful for safety-critical applications, such as healthcare and aerospace software, where rigorous testing is mandatory.

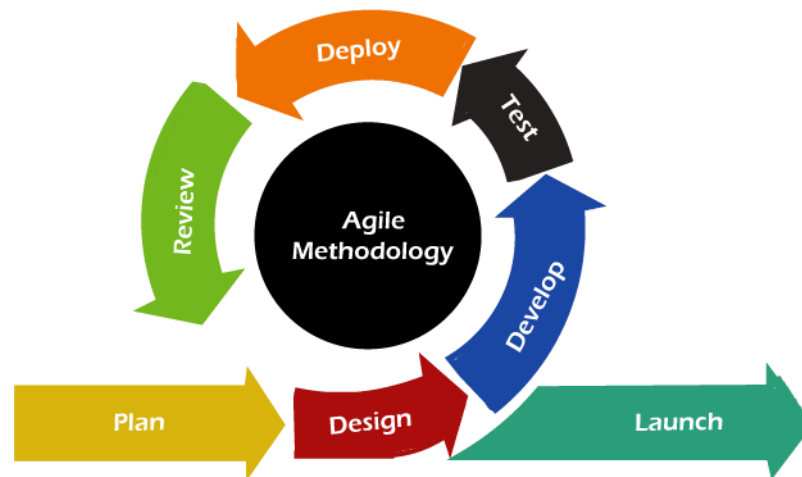


Fig 6.2: Agile Methodology

The Iterative Model focuses on developing an initial version of the software with core functionalities and refining it through multiple iterations based on feedback. This approach reduces risks and ensures that the final product meets user needs. Given the

nature of the project and its requirements, the Agile model is often preferred for projects that demand adaptability, quick delivery, and continuous improvement. Agile's flexibility and focus on customer collaboration make it an ideal choice for modern software development. However, if the project requires extensive risk analysis, the Spiral model may be more suitable. Ultimately, selecting the right software model depends on the project's objectives, complexity, and the level of uncertainty in requirements, ensuring a structured and efficient development process.

6.4 System Architecture

The proposed Smart Intruder Detection System follows a layered architecture to ensure scalability, reliability, and real-time decision support. The system architecture consists of four primary layers: sensing layer, edge processing layer, notification and analytics layer, and application layer.

Sensing Layer

The sensing layer is responsible for capturing real-world data from the environment. It includes PIR motion sensors to detect human movement, audio sensors to capture unusual sounds, and low-light/night-vision cameras to provide continuous visual monitoring in dark conditions. These sensors work together to ensure accurate detection and reliable data collection under various environmental conditions. Processing Module .

Edge Processing Layer

The edge processing layer performs real-time data analysis using a Raspberry Pi. Sensor data is processed locally to reduce latency and maintain data privacy. OpenCV-based skeleton analysis identifies human movement, while deep learning-based face recognition verifies individuals against a secure authorized database. Anomaly detection algorithms detect unusual or suspicious activities, enhancing system intelligence and reducing false alarms.

Notification and Analytics Layer

This layer generates real-time alerts when an intruder or abnormal activity is detected. Alerts containing images or video clips are sent instantly via email or mobile

notifications. The system also logs detected events for further analysis, auditing, and performance improvement.

Monitoring and application layer:

This allows users to receive notifications and view captured images. The email notification system sends real-time alerts with colored captured photos when suspicious activity occurs. This improves security response time and ensures continuous monitoring.

6.5 UML Diagrams

Unified Modeling Language (UML) diagrams are used to visually represent the structure and behavior of the Smart Intruder Detection System. These diagrams help in understanding how different components interact within the system.

The Use Case Diagram shows the interaction between the User and the Intruder Detection System. Main interactions include:

- System monitors the environment using sensors
- Intruder detection through motion, sound, or object movement
- Image capture using USB camera
- Face detection using OpenCV
- Alert generation through LED and buzzer
- Email notification sent to the authorized use

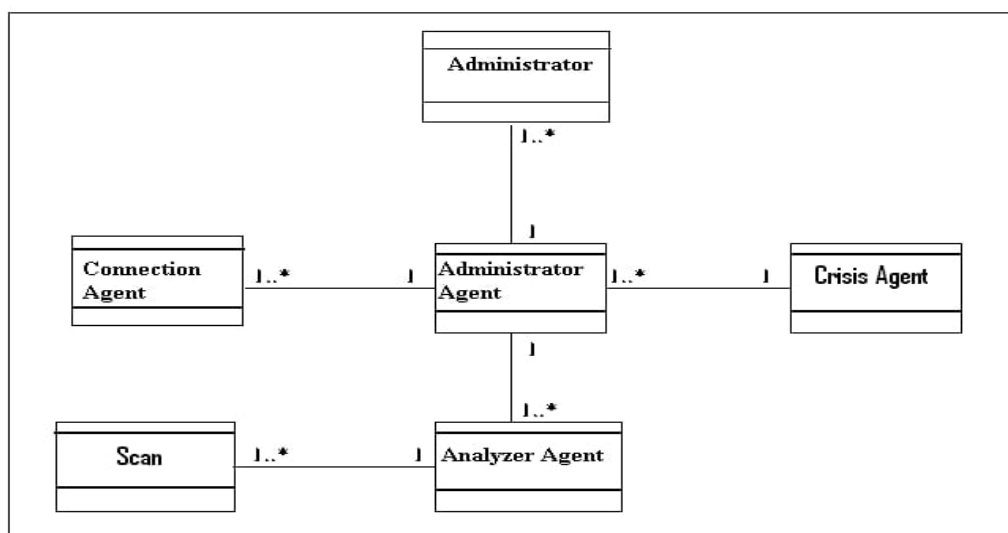


Fig 6.3: Class Diagram

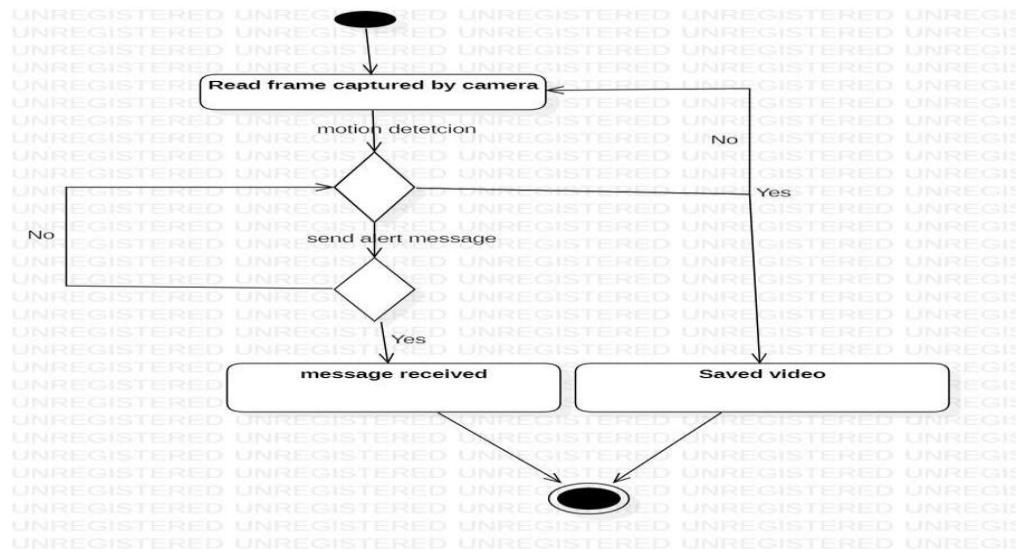


Fig 6.4: Activity Diagram

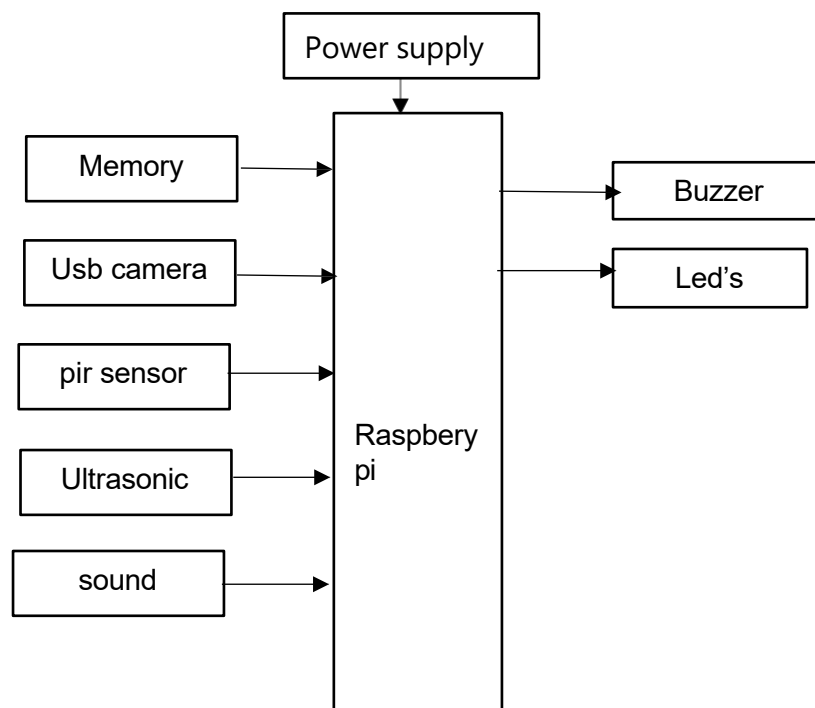


Fig 6.5: Component Diagram

CHAPTER - 7

IMPLEMENTATION

7.1 Introduction

Implementation is an important phase in the development of the Smart Intruder Detection System, where the designed architecture is transformed into a fully functional hardware–software integrated system. This phase involves connecting hardware components such as the Raspberry Pi, PIR motion sensor, ultrasonic sensor, sound sensor, USB camera, LED indicators, and buzzer with the software modules used for processing and alert generation. The Raspberry Pi acts as the central controller of the system. It continuously collects input data from the sensors to monitor motion, sound, and object movement in restricted areas. When suspicious activity is detected, the system activates the USB camera to capture images and uses OpenCV to perform face detection. The captured images are then sent to the authorized user through email notifications, while the system simultaneously activates LED indicators and a buzzer to generate visual and audio alerts. This implementation ensures that the system can detect intruders efficiently, especially in low-light and dark environments, providing a reliable and automated security monitoring solution.

7.2 Technology Used

The development of the Smart Intruder Detection System integrates both hardware and software technologies to achieve real-time intruder detection and alert generation.

Hardware Technologies

- Raspberry Pi (Controller)
- PIR Motion Sensor
- Ultrasonic Sensor
- Sound Sensor
- USB Camera
- LED Indicators
- Buzzer

- Power Supply (5V)

Software Technologies

- Python Programming Language
- OpenCV Library
- Raspberry Pi OS
- SMTP Protocol for sending email notifications
- Arduino IDE
- Visual Studio Code
- Embedded C

The Raspberry Pi processes data received from sensors and activates the camera and alert mechanisms when suspicious activity is detected. Python scripts are used to control the sensors and manage the alert system.

7.2.1 Hardware Technologies

The hardware implementation of the Smart Intruder Detection System is designed to detect unauthorized entry using multiple sensors integrated with the Raspberry Pi. The Raspberry Pi acts as the central processing unit that receives input signals from sensors and processes them. A PIR motion sensor is used to detect human movement in the monitored area. The ultrasonic sensor measures distance changes to detect object movement, while the sound sensor detects unusual sound levels that may indicate suspicious activity. A USB camera is connected to the Raspberry Pi to capture images when motion or abnormal activity is detected. The captured images are processed using the OpenCV library for face detection. For alert generation, LED indicators and a buzzer are connected to the Raspberry Pi. These components provide immediate visual and audio alerts when an intruder is detected. The system is powered using a 5V regulated power supply, ensuring stable operation of the hardware components.

7.2.2 Software Technologies

The software implementation of the Smart Intruder Detection System is developed using Python programming language on the Raspberry Pi platform. Python is used to interface with sensors, control hardware components, process images, and generate alerts. The OpenCV library is used for capturing images from the USB camera

and performing face detection. This helps in identifying intruders and capturing visual evidence during security breaches. The system also uses SMTP (Simple Mail Transfer Protocol) to send email notifications to authorized users when suspicious activity is detected. The email includes the captured colored images of the intruder, allowing the user to quickly respond to potential threats. Development and testing of the software are performed using tools such as Thonny IDE or Visual Studio Code, which provide an efficient environment for Python programming and debugging.

7.2.3 Integration

The Smart Intruder Detection System integrates multiple hardware sensors with software processing modules to provide an automated security solution. The sensors continuously monitor the environment and send data to the Raspberry Pi. When motion, sound, or object movement is detected, the Raspberry Pi processes the sensor data to determine whether suspicious activity has occurred. If an intrusion is detected, the system activates the USB camera to capture images. The captured images are processed using OpenCV for face detection and then stored or sent to the authorized user. At the same time, the system activates LED indicators and a buzzer to generate immediate alerts. The captured images are also sent through email notifications, ensuring that the user receives real-time information about the intrusion. This integration of sensors, camera, and alert mechanisms ensures continuous monitoring and improves the effectiveness of the security system.

7.2.4 Advantages of Software and Hardware Technologies

1. Raspberry Pi

- Provides high processing capability for sensor data and image processing.
- Supports multiple sensor interfaces and camera modules
- Suitable for IoT and embedded security applications.
- Low cost and compact in size.

2. PIR Motion Sensor

- Detects human movement accurately.
- Simple integration with Raspberry Pi.
- Low power consumption.

3. Ultrasonic Sensor

- Measures distance and detects object movement.
- Provides reliable detection of nearby objects.
- Easy interaction with Raspberry Pi.

4. Sound Sensor

- Detects unusual noise levels.
- Helps identify suspicious acoustic activities.

5. USB Camera

- Captures Images of detected intruders.
- Supports face detection using OpenCv.

6. Python Programming Language

- Simple and easy to implement.
- Supports hardware interfacing and automation.
- Large number of libraries available for image processing.

7. OpenCV Library

- Provides advanced image processing capabilities.
- Enables real-time face detection.
- Improves intruder identification.

8. Raspberry Pi OS

- Stable Linux-based operating system.
- Provides strong support for Python and hardware interfacing.

9. SMTP Email Service

- Enables instant alert notification.
- Allows sending captured images to authorized users.
- Improves response time during intrusions.

10. Arduino IDE

- Supports easy sensor integration.
- Provides real-time debugging tools.
- Facilitates efficient hardware programming.

7.3 Pseudocodes

```

1 import RPi.GPIO as GPIO
2 import time
3 import threading
4 import cv2
5 import os
6 from ultralytics import YOLO
7 import smtplib
8 from (module) email import MIMEText
9 from email.mime.multipart import MIMEMultipart
10 from email.mime.image import MIMEImage
11 from datetime import datetime
12
13 # ===== EMAIL CONFIG =====
14 sender_email="224gia3292@srit.ac.in"
15
16 receiver_emails=["234g5a3210@srit.ac.in",
17                  "224gia32b9@srit.ac.in",
18                  "224gia3292@srit.ac.in",
19                  "224gia3290@srit.ac.in", "rajeshbabud.cse@srit.ac.in"]
20
21 sender_password="alqplnjlmgyoxst"
22 smtp_server="smtp.gmail.com"
23 smtp_port=587
24
25 def send_mail(detection, trigger, attachment=None):

```

Fig 7.1:Email Configuration

```

21 3: "Charlie"
22 }
23
24 # Function to detect face and save images for registration
25 def register_face(face_id):
26     cam = cv2.VideoCapture(0)
27     count = 0
28     print("Capturing face images for face ID:", face_id)
29
30     while True:
31         ret, frame = cam.read()
32         gray = cv2.cvtColor(frame, cv2.COLOR_BGR2GRAY)
33         faces = face_cascade.detectMultiScale(gray, 1.3, 5)
34
35         for (x, y, w, h) in faces:
36             count += 1
37             face_img = gray[y:y+h, x:x+w]
38             # Save the captured face in the dataset folder
39             cv2.imwrite(f"{face_dir}/user.{face_id}.{count}.jpg", face_img)
40             cv2.rectangle(frame, (x, y), (x+w, y+h), (255, 0, 0), 2)
41
42     cv2.imshow('Registering Face', frame)

```

Fig 7.2: Face Registration

```

97 # ===== SENSOR THREADS =====
98 def ultrasonic():
99     global distance
100     while True:
101         GPIO.output(TRIG,1);time.sleep(0.00001);GPIO.output(TRIG,0)
102         s=time.time();e=time.time();t=time.time()+0.04
103         while GPIO.input(ECHO)==0:
104             s=time.time()
105             if time.time()-s>t:break
106         while GPIO.input(ECHO)==1:
107             e=time.time()
108             if time.time()-e>t:break
109         with lock:distance=round((e-s)*17150,2)
110         time.sleep(0.3)
111
112 def pir_t():
113     global pir
114     while True:
115         with lock:pir=GPIO.input(PIR)
116         time.sleep(0.1)
117
118 def sound_t():
119     global sound
120     while True:
121         with lock:sound=GPIO.input(SOUND)

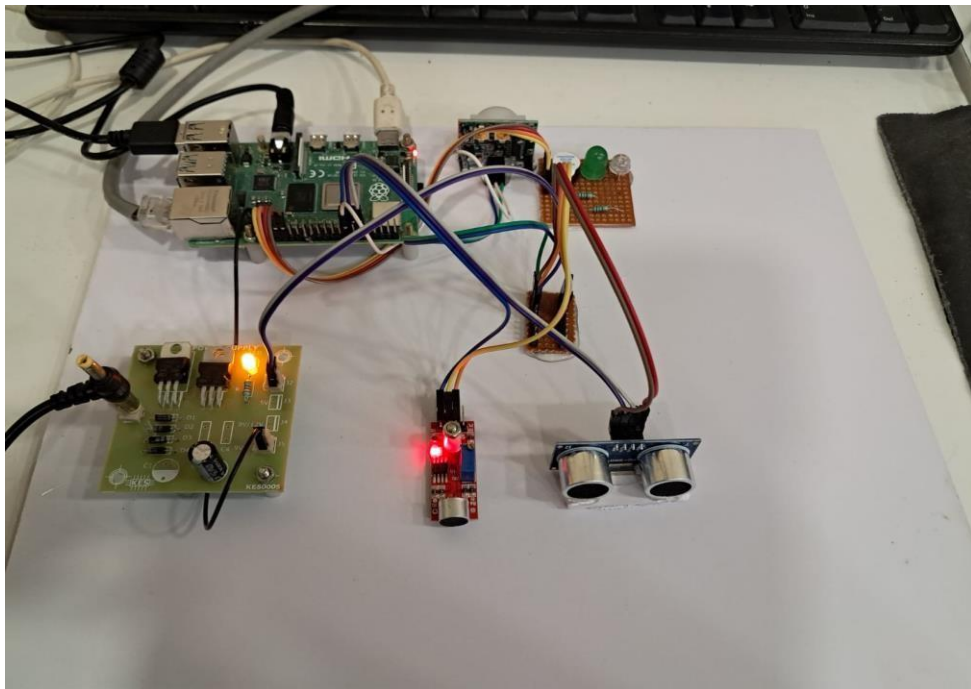
```

Fig 7.3: Sensor Threads

```
128 # ===== CAMERA =====
129 cap=cv2.VideoCapture(0)
130 print("Camera Started")
131 print("PIR Calibrating...")
132 time.sleep(20)
133
134 try:
135     while True:
136         ret,frame=cap.read()
137         if not ret:break
138
139         with lock:d,p,s=distance,pir,sound
140         now=time.time()
141
142         print(f"\r[d]cm PIR:{p} SOUND:{s} COOLDOWN:{pir_cooldown_active}")
143
144         # ----- PIR EDGE DETECTION -----
145         pir_event=False
146         if p==1 and pir_last==0 and not pir_cooldown_active:
147             pir_event=True
148             pir_cooldown_active=True
149             pir_last_alert_time=now
150             print("\r PIR MOTION DETECTED")
151
```

Fig 7.4: Camera

7.4 Hardware

**Fig 7.5: Hardware**

CHAPTER - 8

TESTING

8.1 Introduction

Testing is an essential stage in the development of the Smart Intruder Detection System, as it verifies whether the system operates correctly and reliably before practical deployment. In this phase, both the hardware components and software modules are examined to ensure that the system can accurately detect unauthorized entry and generate alerts.

The system is tested by evaluating the performance of the Raspberry Pi, PIR sensor, ultrasonic sensor, sound sensor, USB camera, LED indicators, and buzzer under different environmental conditions. These tests confirm that the sensors can detect motion, sound, or object movement effectively and that the system can capture images and notify the authorized user. Proper testing improves the stability, accuracy, and reliability of the security system.

8.2 Design of Test Cases and Scenarios

The system is tested by evaluating the performance of the Raspberry Pi, PIR sensor, ultrasonic sensor, sound sensor, USB camera, LED indicators, and buzzer under different environmental conditions. These tests confirm that the sensors can detect motion, sound, or object movement effectively and that the system can capture images and notify the authorized user. Proper testing improves the stability, accuracy, and reliability of the security system.

Different test scenarios are created to observe the system's behaviour in various situations, such as human movement, environmental noise, or object movement. By performing these tests, the system's ability to correctly identify suspicious activities and reduce false alarms is evaluated. This process ensures that the system performs reliably in real-world security environments.

8.2.1 Methods of Testing

Several testing methods are used to evaluate the performance of the Smart Intruder Detection System. Manual testing is carried out to verify system operations such as sensor detection, camera activation, and alert notifications. Hardware testing ensures that sensors and other electronic components are functioning properly when connected to the Raspberry Pi.

Functional testing is conducted to check whether each module performs its intended task. For example, sensors are tested to detect motion or sound, while the camera is tested for capturing images when activity is detected. These testing methods help confirm that all parts of the system work together effectively.

8.2.2 Methodologies of Software Testing

Software testing methods are used to evaluate the performance and reliability of the Python-based control program used in the system. During development, static testing methods such as debugging and code inspection are used to identify logical errors in the program.

Dynamic testing is performed by running the system and observing its behaviour during actual operation. This type of testing verifies that the program correctly reads sensor inputs, activates the camera, and sends alerts when an intrusion occurs. By applying these testing methodologies, the system software becomes more reliable and efficient.

8.2.2.1 Unit Testing

Unit testing focuses on testing individual components of the system separately. Each sensor and module is tested independently to confirm its functionality before integrating it with other components. That each module performs as expected before being integrated with other components.

For instance, the PIR sensor is tested to verify its ability to detect motion, while the ultrasonic sensor is tested to measure distance accurately. The sound sensor is evaluated to detect unusual noise levels. The camera module is also tested

independently to ensure proper image capturing and processing. These individual tests help identify faults early in the development process.

8.2.2.2 Integration Testing

Integration testing is performed after all individual components have been verified through unit testing. In this stage, all hardware and software modules are combined to ensure that they communicate and function correctly as a complete system.

During integration testing, the sensors send input signals to the Raspberry Pi, which processes the data and triggers the appropriate actions such as image capture, alarm activation, and email notification. This testing confirms that the interaction between sensors, processing units, and alert mechanisms works smoothly without errors.

8.3 Testing of Hardware

Hardware testing is carried out to confirm that all electronic components used in the Smart Intruder Detection System function correctly. Each hardware module is tested to ensure proper connectivity and performance.

The Raspberry Pi is tested to verify that it can read sensor data and control output devices such as LEDs and buzzers. The PIR sensor is checked for accurate motion detection, while the ultrasonic sensor is tested for reliable distance measurement. The sound sensor is evaluated to detect abnormal noise levels. The USB camera is tested to confirm that it can capture images when triggered by the system. These tests ensure that the hardware operates efficiently under different conditions.

8.4 Validation

Validation ensures that the Smart Intruder Detection System achieves its intended purpose of providing reliable security monitoring. In this phase, the system is evaluated in real-time conditions to verify its ability to detect intruders and generate alerts.

The validation process confirms that the system can successfully detect motion, capture images of suspicious activity, and notify the authorized user through alerts. It also verifies that the system performs effectively in environments with low lighting or

minimal human supervision. Successful validation demonstrates that the system provides an efficient and automated security solution for restricted areas.

CHAPTER - 9

RESULTS

9.1 Introduction

The results phase evaluates the performance and effectiveness of the Smart Intruder Detection System using Raspberry Pi in detecting unauthorized entry and generating real-time alerts. The system was tested under different environmental conditions, including low-light and dark environments, to verify its reliability. The integrated sensors such as the PIR motion sensor, ultrasonic sensor, and sound sensor successfully detected motion, object movement, unusual sounds, and other abnormal activities in the monitored area. When suspicious activity was detected, the Raspberry Pi activated the USB camera to capture coloured images of the intruder, even during night-time conditions. The system then processed the images and sent them to the authorized user through email notifications, while simultaneously triggering the LED indicators and buzzer for immediate alerts. The results show that the system effectively performs motion detection, sound detection, anomaly detection, and intruder identification, providing a reliable and automated security monitoring solution even in dark environments.

9.2 Output Screenshots

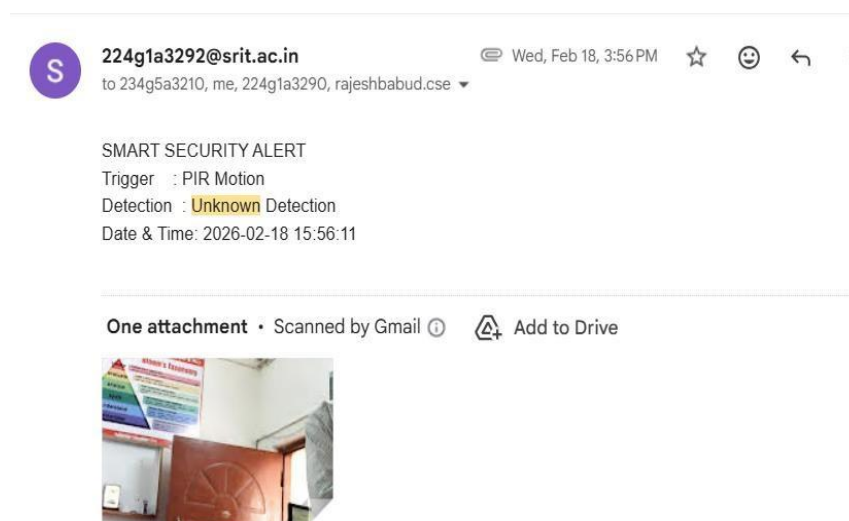


Fig 9.1: Security Alert for unknown motion in day time

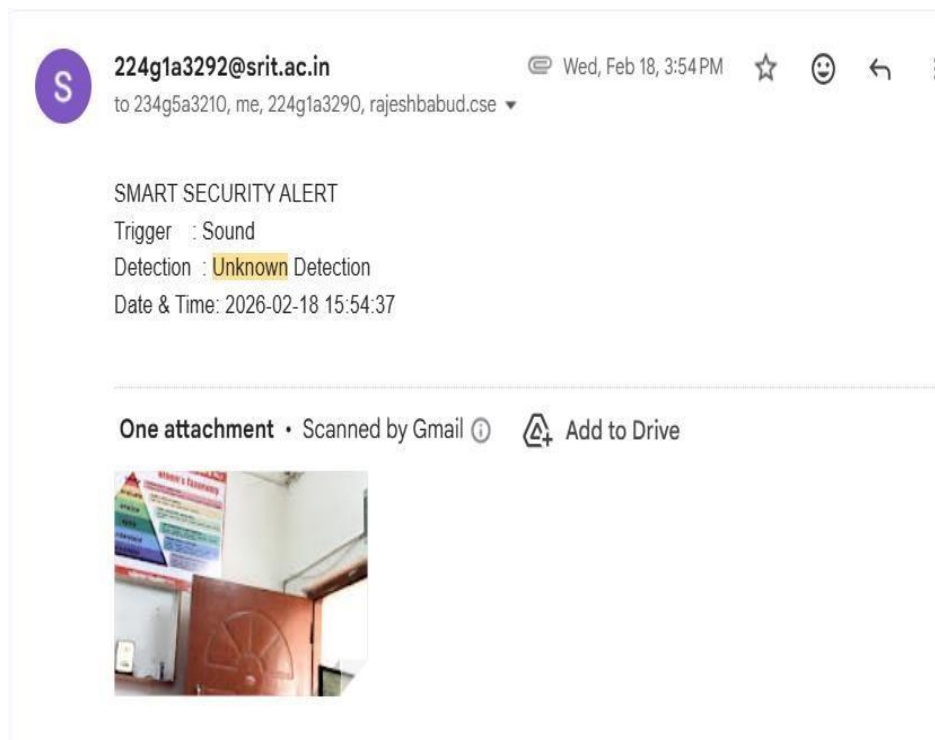


Fig 9.2: Security Alert for unknown Sound in day time

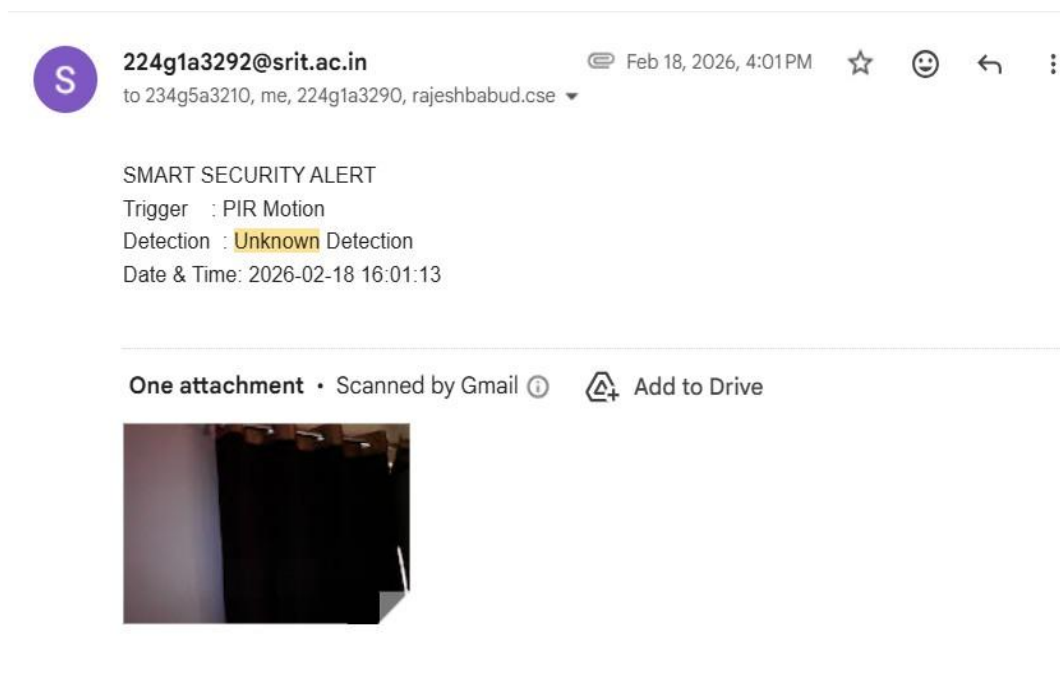


Fig 9.3: Security Alert for unknown motion in dark environment

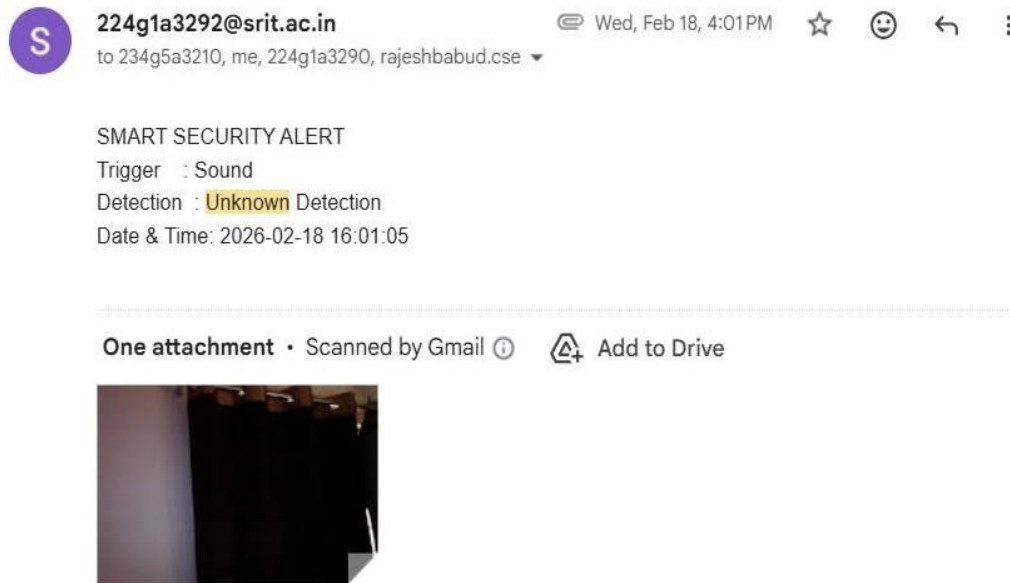


Fig 9.4: Security Alert for unknown sound in dark environment

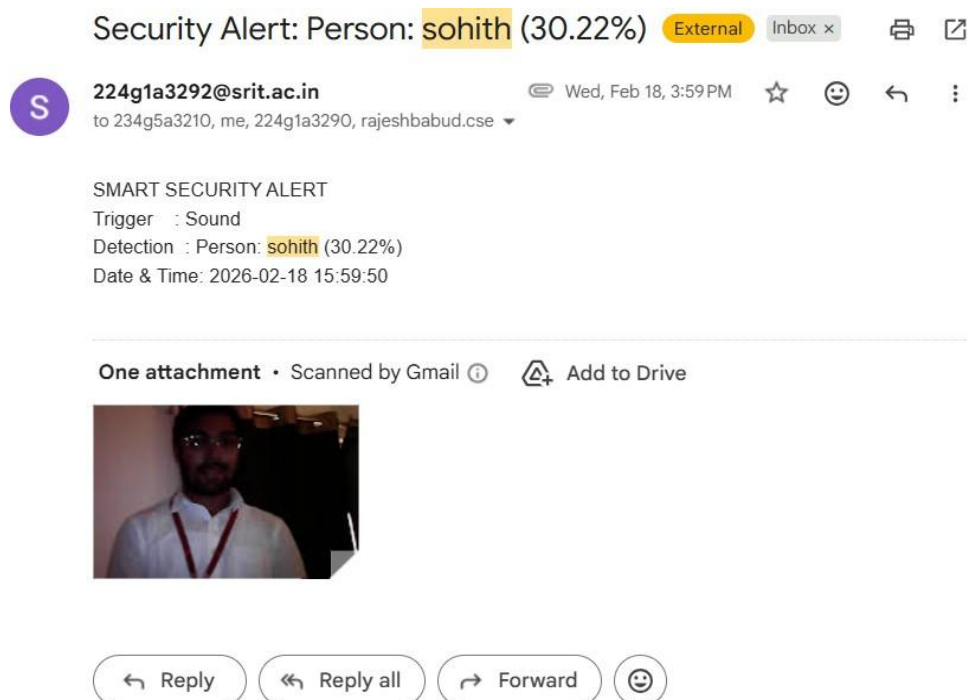


Fig 9.5: Security Alert for known sound in dark environment

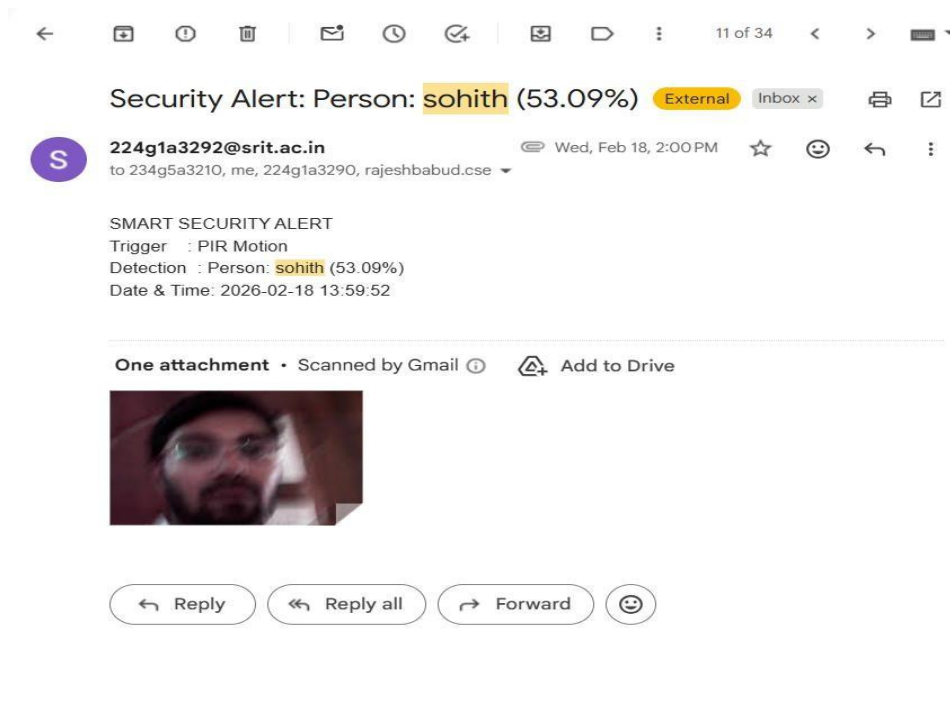


Fig 9.6: Security Alert for known motion in dark environment

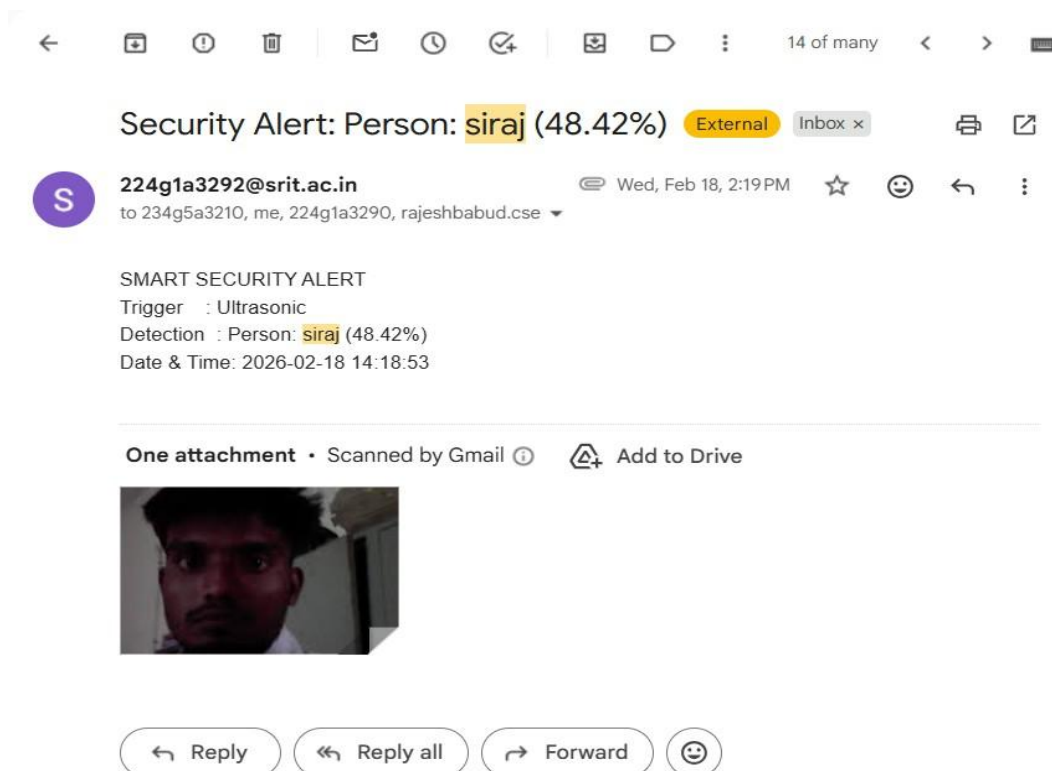


Fig.9.7: Security Alert for known Person in dark environment

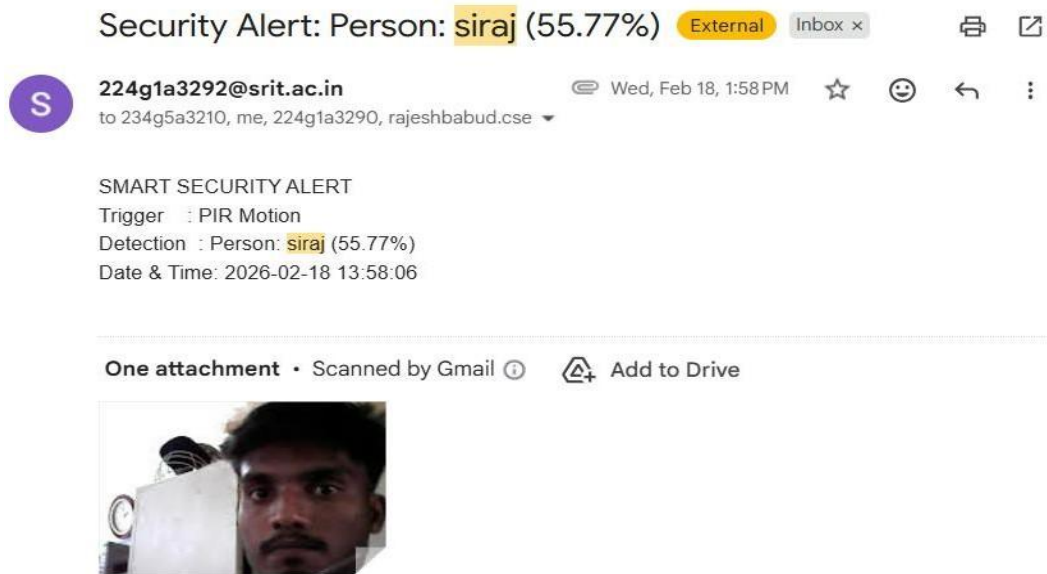


Fig.9.8 Security Alert for known Person in Bright environment

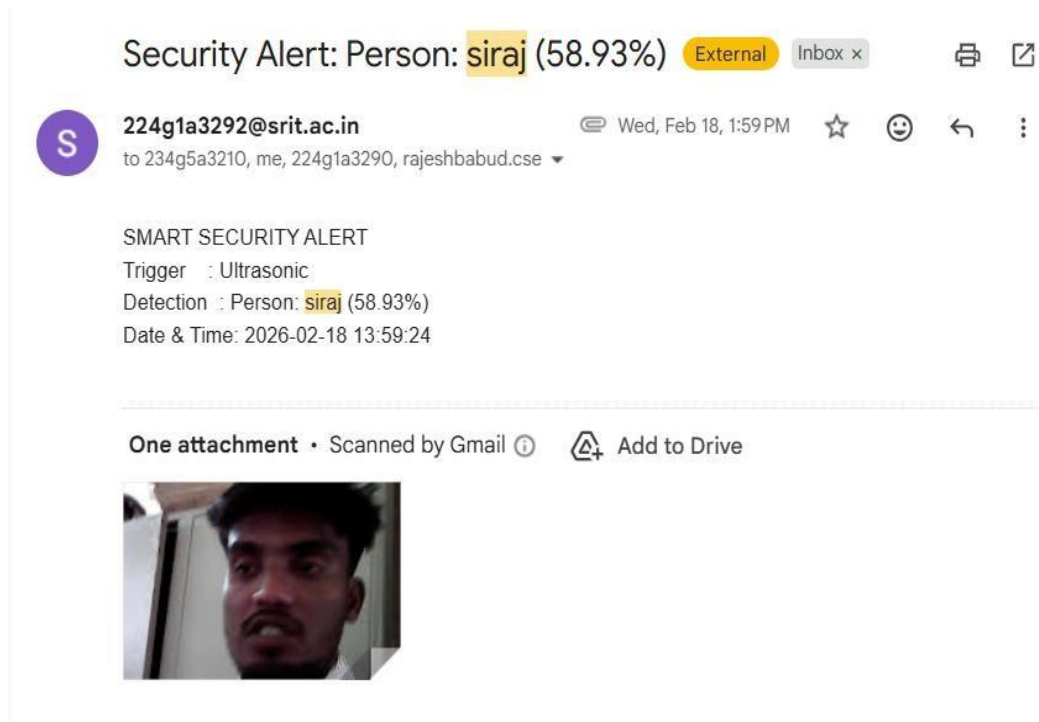


Fig.9.9 Security Alert for known Person in Bright environment

CONCLUSION

The successful implementation of the Smart Intruder Detection System using Raspberry Pi demonstrates the effectiveness of the proposed system in overcoming the limitations of traditional security systems. By integrating multiple sensors such as PIR, ultrasonic, and sound sensors along with image processing using OpenCV, the system provides a more accurate, efficient, and automated solution for intruder detection. The project has been thoroughly tested under various real-world conditions, including low-light and dark environments, ensuring reliable performance. The system is capable of detecting motion, sound, and abnormal activities while capturing coloured images of intruders even during night-time and sending instant alerts to authorized users. This multi-sensor and intelligent detection approach significantly reduces false alarms and improves overall system reliability and response time.

The development process, including analysis, design, implementation, and testing, has ensured that the system meets its intended objectives of providing real-time monitoring, improved accuracy, and enhanced security. The results clearly indicate that combining sensor-based detection with automated alert mechanisms offers better performance compared to conventional surveillance systems. Additionally, the system reduces the need for continuous human monitoring, making it a cost-effective and user-friendly solution.

Despite the successful outcomes, there is scope for further improvement and enhancement. Future work can focus on integrating advanced AI-based face recognition, cloud storage for image data, and mobile application support for real-time notifications. The system can also be expanded to include additional sensors and improved anomaly detection techniques for higher accuracy. Overall, this project establishes a strong foundation for developing intelligent, automated, and scalable security systems, highlighting the importance of IoT and smart technologies in modern surveillance applications.

Future Scope of the Project

The Smart Intruder Detection System provides a reliable and automated solution for security monitoring in restricted areas. However, there are several opportunities to further enhance the system using advanced technologies and features in the future:

- **Integration of Mobile Application:**

A dedicated mobile application can be developed to provide users with real-time notifications, live monitoring, and remote access to the system. This will improve user convenience and enable instant response to security threats.

- **AI-Based Face Recognition and Analysis:**

Advanced Artificial Intelligence and Machine Learning algorithms can be integrated to improve face recognition accuracy and enable intelligent identification of authorized and unauthorized individuals.

- **Advanced Sensor Integration:**

Additional sensors such as thermal sensors, gas sensors, and vibration sensors can be incorporated to enhance detection capabilities and improve system accuracy under different environmental conditions.

- **Cloud Storage Integration:**

The system can be connected to cloud platforms to store captured images and videos securely. This allows remote access, backup, and efficient data management.

- **Scalability with Cloud and Edge Computing:**

Integration of cloud and edge computing technologies can improve system scalability, enable faster data processing, and support real-time decision-making.

- **Integration with Smart Home Systems:**

The system can be integrated with smart home technologies such as smart locks, alarms, and IoT devices to create a complete automated security solution.

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Application Details

APPLICATION NUMBER	202641035910
APPLICATION TYPE	ORDINARY APPLICATION
DATE OF FILING	24/03/2026
APPLICANT NAME	Srinivasa Ramanujan Institute of Technology
TITLE OF INVENTION	IOT-Based Smart Intruder Detection System
FIELD OF INVENTION	ELECTRONICS
E-MAIL (As Per Record)	mail2patentipr@gmail.com
ADDITIONAL-EMAIL (As Per Record)	
E-MAIL (UPDATED Online)	
PRIORITY DATE	
REQUEST FOR EXAMINATION DATE	--
PUBLICATION DATE (U/S 11A)	03/04/2026

Application Status

APPLICATION STATUS

Awaiting Request for Examination

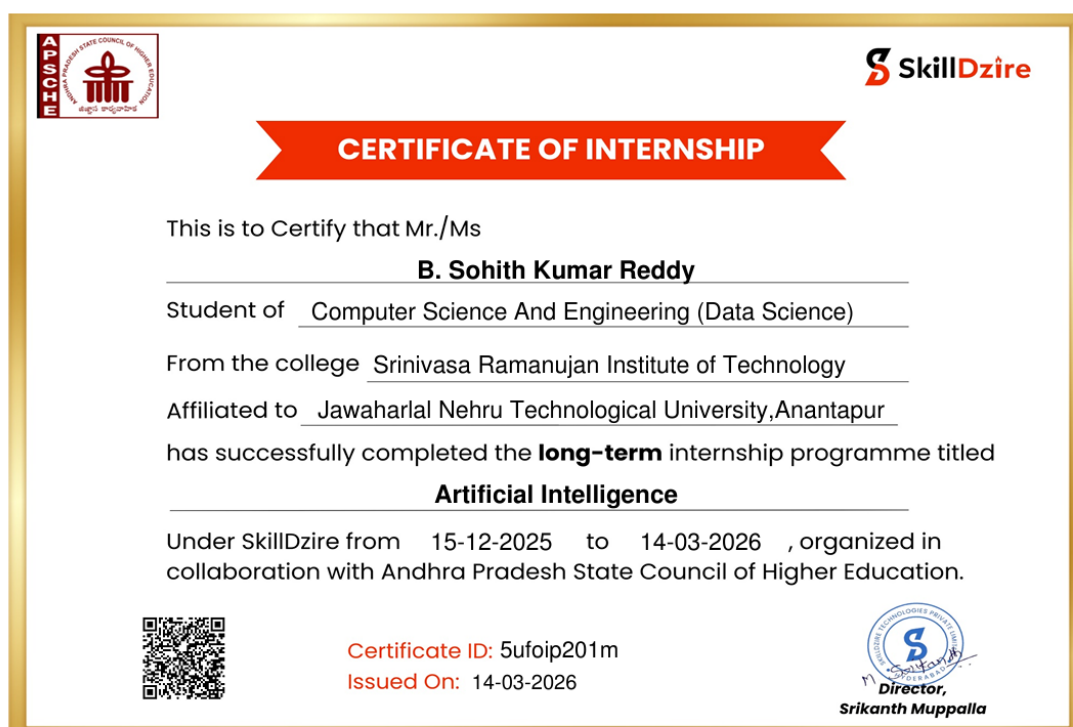
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In case of any discrepancy in status, kindly contact ipo-helpdesk@nic.in

CERTIFICATES







PROJECT CO-PO'S

Course Outcomes (COs)

CO1: Understand the fundamentals of IoT-based security systems and their application in intrusion detection.

CO2: Design and develop an integrated system using Raspberry Pi, motion sensors, and sound sensors for real-time monitoring.

CO3: Implement image capture and basic face recognition techniques for identifying intruders.

CO4: Develop an alert system to send notifications for real-time intrusion detection.

CO5: Analyze sensor data and detection results to reduce false alarms using multi-sensor integration and anomaly detection.

CO6: Evaluate the effectiveness of the system in improving security, reliability, and automated surveillance in real-world environments.

Program Outcomes (POs)

PO 1: Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO 2: Problem Analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO 3: Design / Development Of Solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO 4: Conduct Investigations Of Complex Problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO 5: Modern Tool Usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO 6: The Engineer And Society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO 7: Environment And Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO 10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
CO1	✓	✓	✓	✓	✓	✓	✓	
CO2	✓	✓	✓	✓	✓		✓	
CO3	✓	✓	✓	✓	✓	✓	✓	
CO4	✓	✓	✓	✓	✓	✓	✓	✓
CO5	✓	✓	✓	✓	✓		✓	
CO6	✓	✓	✓	✓	✓	✓	✓	✓

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